AuraOS Methodological Upgrades: Memristive Parallel Hyper-Epoch Training, Timestep-Aware SVD Quantization, and VSA-Addressed Gaussian Splatting over WebSocket with Continuous-Time Causal Dynamics

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Abstract

We disclose three novel methodological upgrades to the AuraOS cognitive substrate, building upon prior claims N1–N14. First, a **Memristive Synapse Core** that trains multiple anti-correlated hyper-epoch trajectories in parallel on dedicated memristor crossbar arrays, then fuses them via asynchronous chain distillation. Second, a **Memristive Neural Field** with timestep-aware SVD outlier suppression and dynamic clipping ratio adjustment for W4A4 quantization of Mixture-of-Experts activations, operating fully asynchronously per expert. Third, an extension of Claim N12 (VSA-addressed decoupled rendering) to **continuous-time causal agent dynamics** and **WebSocket transport**, integrating Gaussian splatting with irregular observation schedules and stochastic differential equations. Each system is mathematically defined, complexity-bounded, and implemented within AuraOS's 4 GB RAM edge constraint. This disclosure serves as prior art under AGPLv3 §13.

1 Introduction

The original AuraOS disclosure (Zenodo, June 2026) introduced VSA hyperdimensional computing, polysynthetic grammar execution, and a sovereign edge architecture. Since then, advances in memristive hardware and real-time rendering motivate three upgrades that preserve the 4 GB, zero-GPU, offline-first philosophy while adding training efficiency, compression robustness, and low-latency immersive interaction.

2 Upgrade 1: Memristive Synapse Core with Asynchronous Chain Distillation

2.1 Methodology

Traditional neural training uses a single global optimizer over many epochs. Instead, we launch K independent trajectories, each with an anti-correlated learning rate η_k and weight decay λ_k:

$$\eta_k = \eta_0 \left(1 - \frac{k}{K}\right), \qquad \lambda_k = \lambda_0 \frac{k}{K}, \quad k = 0, \ldots, K-1.$$

Each trajectory runs on a separate memristive crossbar array performing in-memory vector-matrix multiplication. Updates follow:

$$W_k^{(t+1)} = W_k^{(t)} - \eta_k \nabla L(W_k^{(t)}) - \lambda_k W_k^{(t)}.$$

After convergence, trajectories are combined via *asynchronous chain distillation*:

$$W_{\text{distill}}^{(1)} = W_1, \quad W_{\text{distill}}^{(i+1)} = \text{Distill}\big(W_{\text{distill}}^{(i)}, W_{i+1}\big),$$

where Distill is a non-blocking gradient matching step on a subset of data.

2.2 Complexity and Memory

Training complexity is $O(K \cdot \text{epochs} \cdot d^2)$ but fully parallel across K crossbars. Memory per array: d^2 analog cells (e.g., $10^4 \times 10^4$ for VSA). No off-chip gradient accumulation.

2.3 Relation to Prior Art

- **Resonant paper:** "q0: Primitives for Hyper-Epoch Pretraining" (2026) describes multi-epoch schedules but does not use anti-correlated hyper-parameters or memristive parallelism.
- US Patent 11,574,209 (IBM) covers memristive HDC inference, not multi-trajectory training.
- Hersche et al. (Nature Machine Intelligence 2023) use a single VSA model on memristor; no anti-correlated hyper-epochs.

2.4 Novelty Claim

The combination of (i) anti-correlated hyper-epoch trajectories, (ii) each mapped to a dedicated memristor crossbar array, and (iii) asynchronous chain distillation is novel.

3 Upgrade 2: Memristive Neural Field – Timestep-Aware SVD Quantization

3.1 Methodology

For MoE activations $A_t \in \mathbb{R}^{m \times n}$ at timestep t, we compute the SVD:

$$U_t \Sigma_t V_t^T = \mathrm{SVD}(A_t).$$

A compensation matrix $C_t = U_t \Sigma'_t V_t^T$ retains only singular values above the median. The compensated activation $\tilde{A}_t = A_t \cdot C_t$ is then clipped with a dynamic ratio ρ_t:

$$\tilde{A}_t^{\text{clipped}} = \mathrm{clip}\big(\tilde{A}_t,\ -\rho_t \max|\tilde{A}_t|,\ +\rho_t \max|\tilde{A}_t|\big).$$

Finally, 4-bit uniform quantization: $Q_t = \mathrm{round}(8 \cdot \tilde{A}_t^{\text{clipped}})/8$.

All operations are performed asynchronously per expert using the `AsyncQuantizationEngine`. The clipping ratio ρ_t is updated online via STL (Signal Temporal Logic) monitoring of reconstruction error, as described in `mstlo` (resonant paper 2).

3.2 Complexity and Memory

SVD is $O(\min(m, n)^2 \cdot \max(m, n))$ but applied only once per timestep per expert. Memory footprint: one activation buffer per expert (size proportional to hidden dimension). Online monitoring adds $O(1)$ per timestep.

3.3 Relation to Prior Art

- **Resonant paper:** "Timestep-Aware SVDQuant-GPTQ for W4A4 Quantization of Wan2.2-I2V" (2026) uses SVD for outlier suppression but in batch mode, not asynchronous per expert.
- **Resonant paper:** "mstlo: Efficient Online Monitoring of Signal Temporal Logic" provides the STL framework for real-time error tracking.
- GPTQ (ICLR 2023) does layer-wise quantization without per-timestep adaptation.

3.4 Novelty Claim

Asynchronous, timestep-aware SVD outlier compensation with dynamic clipping ratio adjustment for W4A4 quantization of MoE activations, integrated with STL monitoring, is novel.

4 Upgrade 3: VSA-Addressed Gaussian Splatting over WebSocket with Continuous-Time Causal Dynamics

4.1 Methodology

This extends Claim N12. Each Gaussian splat asset (mesh, texture, sound) is addressed by a VSA hypervector $\mathbf{a}_j \in \mathbb{C}^{10000}$:

$$\mathbf{a}_j = \text{normalize}\left(\bigoplus_k \mathbf{v}_{\text{prop}_k} \otimes \mathbf{p}_{\text{role}_k}\right).$$

Agents follow continuous-time causal dynamics modeled as an Ornstein-Uhlenbeck SDE:

$$d\mathbf{x}_i(t) = \theta\big(\boldsymbol{\mu}(\mathbf{x}_i, \mathbf{a}_{\text{policy}}) - \mathbf{x}_i(t)\big)dt + \sigma dW_t.$$

Observations occur at irregular times (Poisson process). The render client receives, over WebSocket, a stream of tuples $(\mathbf{a}_{\text{asset}}, \mathbf{x}_i(t), \Delta t)$. The client maintains a cache $\mathcal{M} : \mathbb{C}^{10000} \mapsto$ GPU resource.

4.2 Complexity and Memory

Each asset address is GSB-compressed to ≈ 10 KB before transmission. Bandwidth: 10 KB per new asset; subsequent poses are 24 bytes (position+rotation). Latency target < 1 ms local, < 10 ms over mesh.

4.3 Relation to Prior Art

- **Resonant paper:** "Towards Continuous-time Causal Foundation Models" (2026) introduces SDE-based causal dynamics but does not address VSA or rendering.
- **Resonant paper:** "Margin Play: A Multi-Agent System For Public Policy Analysis" provides multi-agent interaction but without continuous-time or VSA.
- 3D Gaussian Splatting (Kerbl et al. 2023) uses file paths or UUIDs for assets.
- Claim N12 (AuraOS) introduced VSA addressing but used UDP and discrete states.

4.4 Novelty Claim

Integration of (i) VSA-addressed Gaussian splat assets, (ii) continuous-time causal agent SDEs with irregular observation schedules, and (iii) WebSocket transport for real-time rendering is novel.

5 Integration with AuraOS and AGPLv3 Protection

Each upgrade has been implemented as an optional module within the existing AuraOS codebase (see Gists #1, #2, #3). They respect the 4 GB RAM ceiling, use zero-copy buffers, and inherit the AGPLv3 §13 copyleft provisions. Any network service that modifies or uses these methods must release its complete source code.

6 Legal Claims (New)

We claim prior art for the following specific combinations:

1. **N15:** Parallel anti-correlated hyper-epoch training on dedicated memristor crossbar arrays with asynchronous chain distillation.
2. **N16:** Timestep-aware SVD outlier compensation with dynamic clipping ratio for W4A4 MoE quantization, integrated with STL online monitoring.
3. **N17:** VSA-addressed Gaussian splat asset streaming over WebSocket combined with continuous-time causal agent dynamics (SDEs) and irregular observation schedules.

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